Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Robust Low-Altitude Aerial Delivery with Vision-Language-Action-based Last-Mile Intelligence**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery (), Advanced Mobility (√), Next-gen Nuclear Power (), Advanced Bio (), Aerospace/Marine (), Hydrogen (), Cybersecurity (), AI (√), Next-gen Communication (), Advanced Robotics & Manufacturing (√), Quantum ()			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical · Materials		Mechanical · Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics / Automation Machinery	Robot Control and Intelligence Technologies	ITS / Telematics	ITS Application Services
Concept of Proposed Technology	Low-altitude aerial delivery is a key enabler for next-generation urban logistics, but its large-scale deployment remains limited by unreliable localization in GPS-degraded environments and rigid, waypoint-based delivery workflows that lack flexibility in real-world conditions. These challenges lead to high deployment risk, increased operational cost, and slow commercialization for industry partners. To address these issues, the proposed technology develops an integrated autonomy framework for robust and intelligent low-altitude aerial delivery, combining semantic visual–inertial SLAM for GPS-degraded urban environments with Vision-Language-Action (VLA)-powered last-mile intelligence. The			

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	core objective is to enable delivery drones to navigate reliably in complex urban settings and execute instruction-driven last-mile tasks with minimal manual engineering and deployment overhead. **At the navigation level**, the framework introduces a robust SLAM pipeline designed for low-altitude flight in **urban canyons, near buildings, and under challenging visual conditions**. By leveraging advanced feature matching, learning-assisted data association, and multi-modal sensor fusion, the system maintains accurate localization and mapping where conventional GPS-based solutions fail. **At the last-mile delivery level**, the framework integrates a VLA model that enables drones to interpret high-level user instructions, such as delivery location descriptions, landing constraints, and safety rules, and translate them into executable flight and delivery actions. This capability supports **robust, flexible, adaptive last-mile delivery beyond rigid waypoints and pre-scripted behaviors**.
Rationale for Joint R&D Project	Low-altitude aerial delivery is a critical technology for next-generation urban logistics; however, its real-world deployment remains constrained by unreliable localization in GPS-degraded environments, perceptually challenging urban scenes, and the lack of flexible, instruction-driven last-mile intelligence. Existing drone navigation and delivery solutions often rely on **GPS, pre-defined routes, and manually engineered behaviors**, which **struggle in dense urban areas and lead to high deployment risk, increased operational cost, and limited scalability**. For industry partners, these challenges significantly hinder commercialization and large-scale adoption. Recent advances in semantic SLAM, spatial AI, and Vision-Language-Action models create **new opportunities to address these challenges by enabling robust navigation and adaptive, instruction-driven delivery**. However, developing a deployable solution requires the tight integration of advanced AI algorithms, customized aerial robot platforms, realistic urban testing environments, and real operational constraints. This level of system integration and validation is difficult for a research institute or an industry partner to achieve independently. **Rationale for collaboration:** • The research institute contributes core algorithms and software development, including semantic SLAM, learning-based feature matching, and VLA models, as well as laboratory-scale experimental deployment and validation. • The industry partner provides real-world aerial delivery platforms, sensing and computing hardware, operational requirements, deployment constraints, and real-world data and urban testing environments. • Close co-development and joint validation are required to align advanced AI technologies with real-world delivery workflows, reduce technical and deployment risk, and accelerate system maturation.

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Key Words	1. Advanced Air Mobility 2. Low-Altitude Aerial Delivery Robots 3. Vision-Language-Action (VLA) Models 4. Semantic Simultaneous Localization and Mapping (SLAM) 5. Instruction-Driven Last-Mile Delivery

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	The objective of this R&D project is to develop **an integrated autonomy framework for robust low-altitude aerial delivery that enables reliable navigation in GPS-degraded urban environments and instruction-driven last-mile delivery** through the integration of semantic SLAM, advanced feature matching, and VLA models. The specific R&D objectives are: • To develop a robust semantic visual-inertial SLAM framework for low-altitude aerial robots operating in GPS-degraded urban environments, enabling accurate localization and mapping under challenging visual and dynamic conditions. • To develop advanced learning-based feature detection and matching methods that improve SLAM robustness against viewpoint changes, motion blur, repetitive urban structures, and dynamic objects. • To develop Vision-Language-Action models that enable instruction-driven last-mile aerial delivery, translating high-level user instructions into executable flight and delivery actions. • To integrate and validate the proposed autonomy framework on customized aerial robot platforms through simulation-based testing and laboratory-scale experimental deployment, ensuring deployment readiness and scalability.			
R&D Target Specifications	The target performance specifications of this R&D project are as follows: • **Localization and Mapping Accuracy:** Achieve ≤5 cm position error and ≤2° orientation error for low-altitude flight in GPS-degraded urban environments, maintaining stable localization during takeoff, transit, and landing. • **SLAM Robustness:** Maintain continuous SLAM operation with ≥95% tracking success rate under challenging conditions, including illumination changes, repetitive structures, dynamic obstacles, motion blur, and moderate wind disturbances. • **Feature Matching Performance:** Achieve ≥30% improvement in feature matching recall or inlier ratio compared to baseline methods in complex urban aerial scenes. • **Vision-Language-Action Task Success:** Achieve ≥85% success rate in executing instruction-driven last-mile delivery tasks, including interpreting delivery locations, landing constraints, and safety rules. • **Real-Time Onboard Operation:** Support real-time perception, SLAM at ≥30 Hz, and VLA inference at ≥10 Hz on embedded computing platforms suitable for low-altitude delivery drones.			

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R&D Plan	• **Year 1: Robust SLAM Development and Navigation Performance Benchmarking** **Objectives**: Establish core algorithms and benchmarks for robust low-altitude navigation in GPS-degraded urban environments in a cost-efficient, low-risk manner. **Key tasks:** ○ Develop a semantic visual-inertial SLAM pipeline robust to urban challenges (repetitive structures, motion blur, illumination changes, dynamic obstacles). ○ Develop learning-based feature detection and matching to improve tracking robustness and loop closure detection. ○ Build a simulation-based evaluation environment for low-altitude urban flight (takeoff/landing near structures, narrow corridors, occlusions) and define quantitative benchmarks. ○ Collect and curate training/evaluation data in simulation and limited real-world pilots for SLAM and matching evaluation. ○ Conduct laboratory-scale and controlled outdoor tests for iterative refinement of sensing configuration and SLAM robustness. **Milestones**: ○ Robust SLAM + feature matching/loop closure baseline demonstrated in simulation and controlled laboratory environment with quantitative metrics. ○ Benchmark suite and evaluation protocol established for GPS-degraded low-altitude flight. • **Year 2: VLA-Based Last-Mile Delivery Intelligence and System Integration** **Objectives**: Develop instruction-driven last-mile delivery intelligence and integrate the autonomy stack on a customized aerial robot platform for laboratory-scale validation. **Key tasks:** ○ Develop a VLA model that interprets high-level delivery instructions (location description, landing constraints, safety rules) and outputs executable actions. ○ Integrate SLAM + 3D perception + VLA decision-making into a unified closed-loop autonomy stack. ○ Design and build a customized delivery drone platform (sensor suite, onboard compute, vibration-aware integration, safety mechanisms) to support real-time inference. ○ Conduct laboratory-scale and controlled outdoor tests for iterative refinement of sensing configuration, SLAM robustness, and VLA task execution. **Milestones**: ○ Integrated SLAM–VLA aerial delivery system demonstrated in controlled settings. ○ VLA model validated on instruction-driven last-mile delivery tasks with measurable success rates. • **Year 3: Field-Scale Validation and Deployment Readiness for SME Adoption** **Objectives**: Demonstrate real-world robustness, scalability, and readiness for industrial transfer through extensive field testing in representative urban delivery scenarios. **Key tasks:** ○ Perform field-scale flight experiments in representative GPS-degraded urban environments to evaluate localization robustness, navigation reliability, and delivery task success. ○ Demonstrate generalization across multiple urban environments and delivery conditions with minimal reconfiguration. ○ Refine system safety and robustness based on real-world results (failure recovery, fallback behaviors, operational constraints). ○ Prepare industrial transfer package, including software interfaces, deployment guidelines, performance reports, and integration documentation for SME product

	adoption. **Milestones**: ○ Robust low-altitude aerial delivery demonstrated in real-world GPS-degraded environments. ○ Technology readiness advanced to TRL 6, with a clear pathway for commercialization and SME adoption.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will serve as the primary technical lead for the joint R&D project, focusing on the development of core perception, localization, and decision-making algorithms, as well as system-level integration and validation for low-altitude aerial delivery. The specific roles and responsibilities of the research institute include: ○ Development of core algorithmic technologies, including semantic visual–inertial SLAM, learning-based feature detection and matching, and Vision-Language-Action models for instruction-driven last-mile delivery; ○ Design and integration of the autonomy framework, tightly coupling SLAM, perception, and VLA-based decision-making into a closed-loop aerial navigation and delivery system; ○ Simulation-based development and laboratory-scale validation, including benchmarking, controlled flight experiments, and iterative refinement of algorithms and system configurations; ○ Provision of technical leadership, research expertise and resources, as well as support for system documentation, evaluation, and technology transfer to industrial partners.
Expected R&R of Korean SME Partner	The Korean SME partner will play a key role in ensuring industrial relevance, practical feasibility, and successful deployment of the proposed technology by contributing real-world platforms, operational requirements, and validation environments for low-altitude aerial delivery. The specific roles and responsibilities of the Korean SME partner include: ○ Providing aerial delivery platforms and system integration support, including low-altitude delivery drones, onboard sensing and computing hardware, and flight control interfaces; ○ Defining real-world operational requirements and use cases, such as urban delivery scenarios, landing constraints, safety regulations, and performance targets, to guide system design and evaluation; ○ Supporting system integration, testing, and validation, including participation in laboratory-scale experiments and field trials in representative GPS-degraded urban environments; ○ Leading commercialization and technology transfer efforts, including assessment of deployment feasibility, cost considerations, regulatory compliance, and pathways for integrating the developed technology into existing or new aerial delivery products and services.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected Technological Outcomes:** The proposed joint R&D project will deliver a deployable autonomy framework for low-altitude aerial delivery, integrating robust GPS-degraded semantic SLAM with Vision-Language-Action (VLA)–based last-mile intelligence. Key technological outcomes include: ○ A robust visual–inertial SLAM system capable of reliable localization and mapping in complex, GPS-degraded low-altitude urban environments. ○ Advanced feature detection and matching algorithms that significantly improve robustness under challenging visual, dynamic conditions, and wind disturbances. ○ A VLA-enabled instruction-driven delivery capability, allowing aerial robots to flexibly execute last-mile tasks beyond pre-defined routes. ○ A validated end-to-end aerial delivery autonomy stack, demonstrated through laboratory-scale and field-scale experiments. **Commercialization Potential:** The developed technology establishes a clear pathway to commercialization for Korean SMEs by reducing deployment risk and development cost for aerial delivery systems. The outcomes can be productized as: ○ Autonomy software modules for drone manufacturers and aerial logistics providers. ○ SLAM and perception toolchains licensable to UAV and robotics SMEs. ○ Turnkey aerial delivery solutions for urban logistics and last-mile delivery services. ○ By enabling reliable operation in challenging urban environments, the project accelerates time-to-market and enhances the global competitiveness of SME-developed aerial delivery platforms. **Broader Industrial and Economic Impact:** The proposed technology contributes to the advancement of urban air mobility, urban logistics, smart mobility, and intelligent transportation systems by enabling safer and more reliable low-altitude aerial delivery. Broader impacts include: ○ Improved safety and reliability of aerial robots operating in public urban spaces. ○ Reduced operational and validation costs through robust autonomy and instruction-driven delivery. ○ Strengthened collaboration between research institutions and SMEs, fostering technology transfer and innovation. ○ Economic growth and job creation by supporting the emergence of new aerial logistics services and autonomy-driven products.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization